Saurabh Yadav

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Education

Koneru Lakshmaiah University **Guntur, Andhra Pradesh, India**
Bachelor of Technology in Computer Science and Engineering | CGPA: 9.61/10.0 *Aug 2023 – May 2027*

- **Specialization:** AI-Driven Language Technology

Technical Skills

Languages: Python, Java, SQL

AI & Machine Learning: Machine Learning Algorithms, Deep Learning, Neural Networks, Transformers, SpaCy, Scikit-learn, TensorFlow, Keras, PyTorch, Feature Engineering, Model Fine-Tuning

Generative AI & LLMs: Large Language Models (LLMs), RAG (Retrieval-Augmented Generation), Fine-Tuning (LoRA/QLoRA), Prompt Engineering, Vector Embeddings, Quantization, AST Parsing

Backend : FastAPI, REST APIs, Microservices Architecture, SQLAlchemy (ORM), JWT Authentication

Databases & Vector Stores: Vector DBs (ChromaDB, FAISS), PostgreSQL, MySQL

Cloud & MLOps / Tools: Amazon SageMaker, OpenRouter, Git, GitHub, Docker, React.js

Projects

ToxiGuard AI – Real-Time Abuse & Toxicity Moderation Platform **Jul 2026**
Tech Stack: *Python, PyTorch, FastAPI, DeBERTa-v3, ONNX Runtime, WebSockets, React 18, OpenRouter* GitHub | Live Link

- Architected a **3-tier cascaded AI inference pipeline** (RegEx rules → ONNX DeBERTa → LLM reasoning), achieving **94.2% ROC-AUC** across 6 toxicity categories while cutting LLM token costs by **65%**.
- Quantized PyTorch Transformer weights via **ONNX Runtime** CPU optimization, slashing inference latency from 280ms to **<35ms** (**87% speedup**) with negligible model drift ($< 10^{-4}$).
- Engineered a **Manifest V3 Chrome Extension** using debounced **MutationObservers** across 8 social platforms (X, Reddit, IG, LinkedIn), reducing CPU thrashing by **45%** and manual review workload by **70%**.
- Deployed a production **FastAPI REST/WebSocket API** with JWT auth and SlowAPI rate limiting, powering a **React 18** dashboard with perturbation-based **XAI token attribution** and batch CSV processing.

Rigel AI – Intelligent Code Smell Detection Platform **Feb 2026**
Tech Stack: *Python, FastAPI, AST Parsing, Scikit-learn, LLMs, REST APIs, React.js, OpenRouter* GitHub | Live Link

- Engineered a hybrid **AST-driven static analyzer** and **LLM inference engine** to detect **15+ code anti-patterns**, improving detection precision to **94.2%** over standalone LLM baselines.
- Built asynchronous **FastAPI REST endpoints** with UUID-indexed session persistence for multi-turn AI code discussions, reducing multi-file processing latency by **42%** (**<150ms** response time).
- Evaluated ML classifiers on Microsoft's **CodeXGLUE benchmark** and implemented a deterministic **AST syntax validator** (`ast.parse`), reducing hallucinated refactoring suggestions to **<1%**.

Experience

AWS Academy – EduSkills Foundation **Verify**
Artificial Intelligence and Machine Learning Intern (Cohort 12) *Apr 2025 – Jun 2025*

- Designed and deployed supervised learning models on **AWS SageMaker**, attaining **91%+ accuracy** on sentiment analysis (10k+ E-Commerce reviews), **88%+** on sonar predictions (SVM/RF), and **94%+** on **MNIST Digit Classification** (using 3-layer NN).
- Optimized model convergence using **SageMaker Hyperparameter Tuning (HPO)** and k-fold cross-validation, boosting cross-validation metrics across classifiers by **15%**.
- Containerized and published model endpoints utilizing **TensorFlow, Scikit-learn, and Pandas**, implementing automated logging to monitor model performance.

Certifications

- **Microsoft Certified:** Azure AI Apps and Agents Developer Associate (Microsoft, Aug 2026) Verify

Achievements & Activities

- Solved **500+ DSA problems** across LeetCode, HackerRank, CodeChef, and Codeforces.
- **Academic Excellence:** Ranked in the **Top 1%** of the Computer Science department with a cumulative CGPA of **9.61/10**.
- **Hackathon Finalist:** Led a team of 4 at **Code4Change Hackathon** to design and deploy a full-stack ride-booking platform within 24 hours, placing in the **Top 10 out of all competing teams**.
- **Leadership & Sports:** Represented the **Lucknow District U-19 Cricket Team** (2022–2024), demonstrating high-pressure decision making, resilience, and teamwork in competitive sports.